Exploratory Financial Analysis Report

CS Vision – Data Analytics Internship Program (Set A)

Project: Financial & Operational Performance Analytics System

Week 2: Exploratory Financial Data Analysis (27 October – 2 November 2026)

Prepared by: Tanisha Sinha

1. Objective

This report explores the cleaned Global Superstore dataset from Week 1 to understand revenue, expenses, profit, product and regional performance, monthly trends and the relationships between key variables. The findings prepare the ground for the KPI and profitability work in Week 3.

2. Data and Method

Dataset: the clean analytical dataset created in Week 1, with 51,290 order lines and 40 columns, covering 1 January 2011 to 31 December 2014. It includes 25,035 orders, 1,590 customers, 147 countries and 13 regions.

Method: descriptive statistics, grouped aggregation and visual analysis using Python, Pandas, NumPy, Matplotlib and Seaborn. The work is in the notebook Week2_EDA.ipynb, and all 15 charts are saved in the visualizations folder.

Definitions: Gross Profit is the Profit recorded in the data. Cost is Sales minus Profit. Net Profit is Profit minus Shipping Cost, which is treated as the operating expense. Net Margin is Net Profit divided by Sales.

3. Headline Figures

Measure	Value
Total sales	12,642,502
Total cost	11,175,045
Gross profit	1,467,457 (11.6% of sales)
Shipping expense	1,352,821 (10.7% of sales)
Net profit	114,637 (0.91% of sales)
Average sales per order line	246.49 (median 85.05)
Loss-making order lines	21,150 (41.2%)

The business earns a healthy gross margin, but shipping costs absorb almost all of it.

4. Time Trends (Charts 1–4)

Sales growth. Sales grew every year, from 2.26M in 2011 to 4.30M in 2014. Year-on-year growth was 18.5% in 2012, 27.2% in 2013 and 26.3% in 2014.

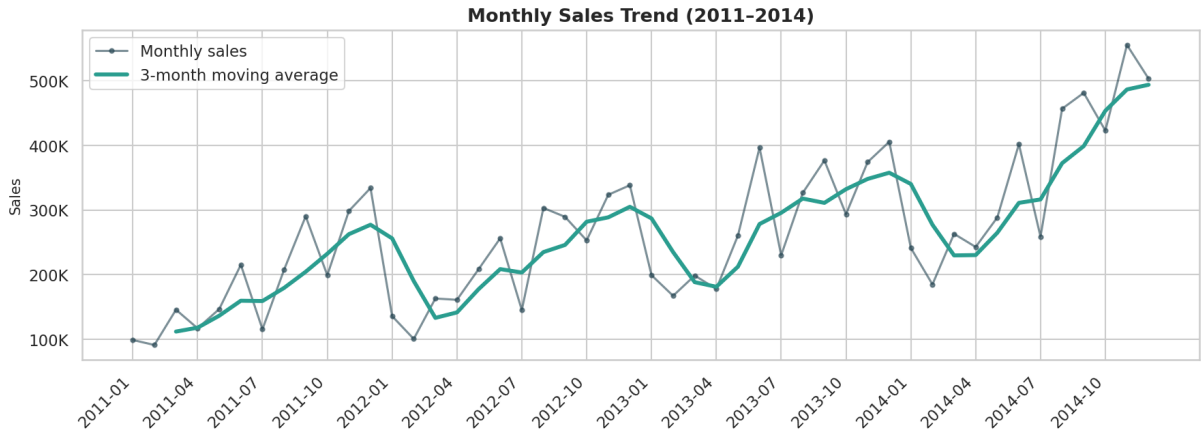


Figure 1: Monthly sales trend with 3-month moving average

Profit has not kept pace. Net margin was 0.21% in 2011, 0.89% in 2012, 1.24% in 2013 and 1.02% in 2014. Net profit rose from about 4.7K to 43.7K, but 15 of the 48 months made a net loss, and December 2014 was the worst month (a loss of about 7.9K).

Seasonality. December, November and September are the strongest months on average, and February, January and April are the weakest. The highest single month was November 2014, with sales of about 555K. February 2011 was the lowest, at about 91K.

5. Product and Category Performance (Charts 5–7)

Category	Sales	Net profit	Net margin
Technology	4.74M	156,730	3.3%
Furniture	4.11M	-155,116	-3.8%
Office Supplies	3.79M	113,023	3.0%

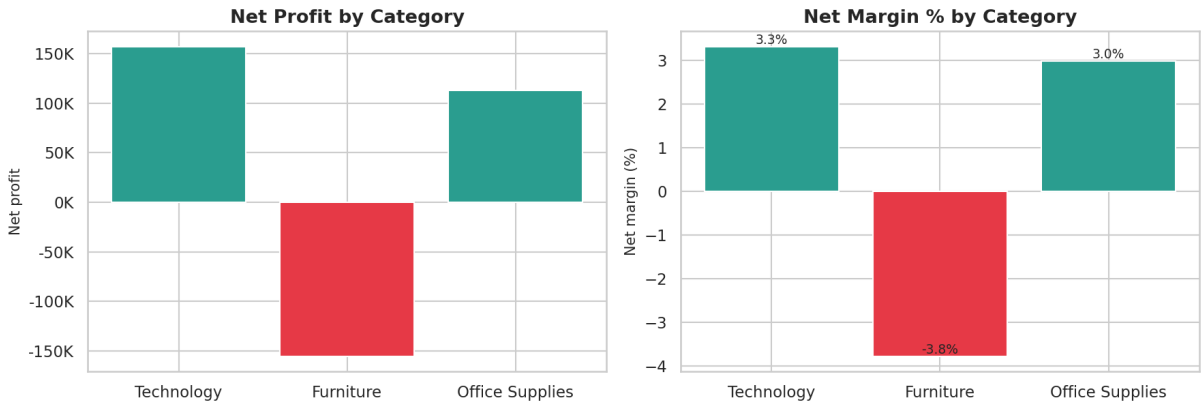


Figure 2: Net profit and net margin by category

Furniture is the only loss-making category. The main reason is **Tables**, which lose about 144K (a -19% margin) on 757K of sales. Chairs (-24K), Machines (-20K), Storage (-12K) and Supplies (-2K) are also negative. The most profitable sub-categories are Copiers (99K), Accessories (46K), Appliances (33K), Paper (33K) and Phones (32K).

At product level, the Canon imageCLASS 2200 Advanced Copier earns the most (about 23.9K). The biggest losers include the Cubify CubeX 3D Printer Double Head Print (about -9.6K) and the Motorola Smart Phone, Cordless (about -8.2K).

6. Regional and Segment Performance (Charts 8–10)

Loss-making regions: Southeast Asia (net loss about 75.8K, margin -8.6%), EMEA (-44.5K, -5.5%) and South (-28.1K, -1.8%). The Caribbean and Oceania are almost break-even.

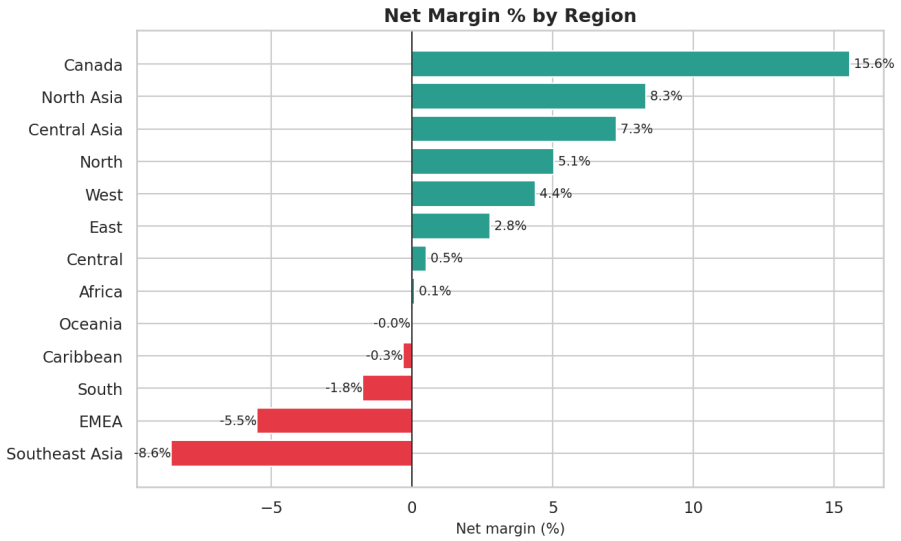


Figure 3: Net margin by region

Best regions by margin: North Asia (8.3%), Central Asia (7.3%) and North (5.1%). Canada has the highest margin (15.6%) but is very small, with 67K of sales.

Largest region: Central contributes 22% of all sales but earns only a 0.5% net margin, so it offers a large opportunity to improve.

Customer segments: Consumer (6.51M sales), Corporate (3.82M) and Home Office (2.31M) have similar margins. Home Office is slightly better at 1.4%, against 0.8% for the other two.

7. Discounts and Shipping (Charts 11–12)

Discounts. Order lines with no discount earn an average net profit of +35 per line. Small discounts (1–10%) still earn about +27. From 11–20% the average falls to about -3, and above that it is clearly negative: -64 (21–30%), -81 (31–50%) and -109 (above 50%). Lines discounted above 20% make up 22.1% of all lines and lose about 1.02M in total.

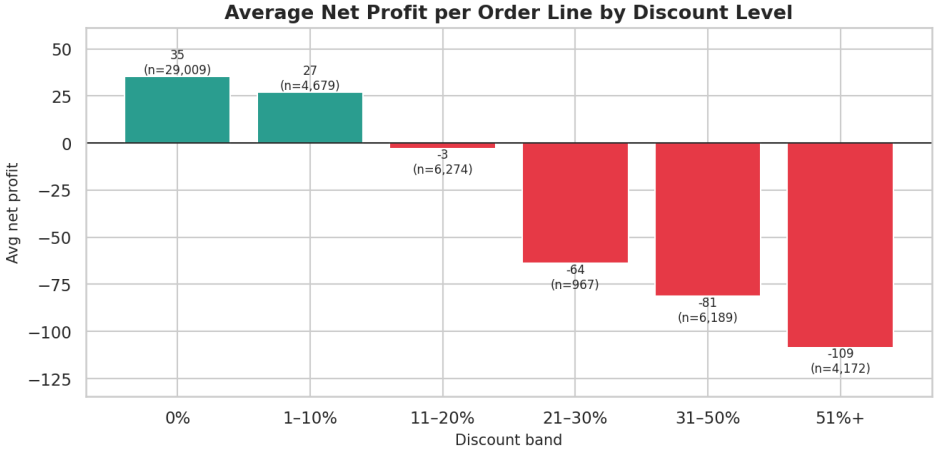


Figure 4: Average net profit per order line by discount level

Shipping. Shipping cost is 10.7% of sales overall. It varies sharply by ship mode: Same Day 17.4%, First Class 16.8%, Second Class 12.2% and Standard Class 8.1%.

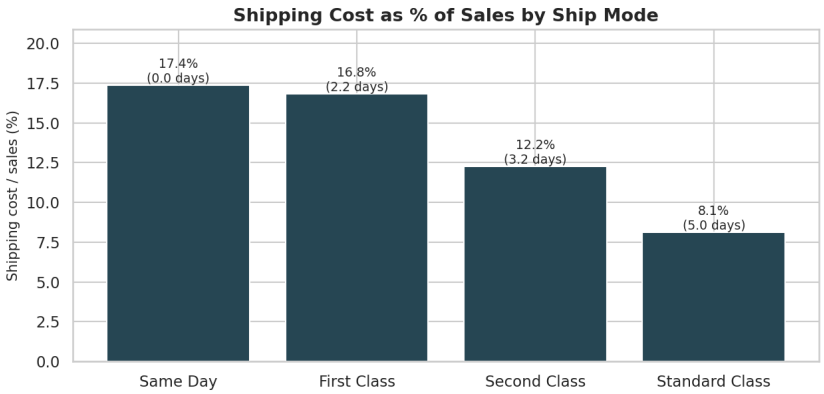


Figure 5: Shipping cost as a percentage of sales, by ship mode

8. Distributions and Relationships (Charts 13–15)

Skewed sales. Order-line sales are strongly right-skewed (median 85, mean 246). The top 10% of lines produce 53.6% of total sales.

Profit distribution. Most lines make a small profit or loss, while a few large losses pull the total down. 41.2% of lines lose money after shipping.

Correlations. Discount and Profit are negatively correlated (r = -0.32), meaning higher discounts go with lower profit. Sales and Shipping Cost are strongly positively correlated (r = 0.77), and Sales and Profit moderately so (r = 0.49).

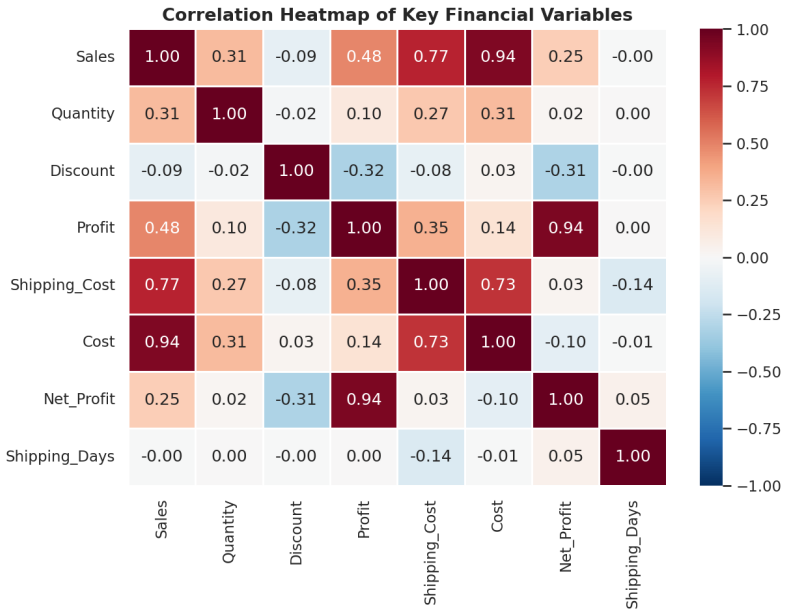


Figure 6: Correlation heatmap of key financial variables

9. Key Insights

1. Sales have more than doubled in four years, but net margin has stayed at about 1%.
2. Shipping costs, at 10.7% of sales, remove almost the whole gross margin.
3. Discounts above 20% are the biggest single cause of loss, at about -1.02M.
4. Furniture, and Tables in particular, loses money despite strong sales.
5. Southeast Asia, EMEA and South are loss-making regions.
6. Same Day and First Class shipping cost about twice as much, relative to sales, as Standard Class.
7. Q4 (September to December) is the peak season.
8. About half of all sales come from the top 10% of order lines.

9. Technology, particularly Copiers, is the most reliable profit source.
10. Central is the biggest region but earns almost no profit.

10. Recommendations

- Cap discounts at about 20% and require approval above that level.
- Review the pricing and discounting of Tables, Chairs, Storage and Machines.
- Investigate shipping costs and pricing in Southeast Asia, EMEA and South.
- Encourage Standard Class shipping and charge a premium for Same Day and First Class.
- Plan inventory and promotions around the September–December peak.
- Grow Technology and Copiers, which have the strongest margins.
- Improve profitability in Central, where the sales base is already large.

11. Assumptions and Limitations

- Cost and Net Profit are derived. If the source Profit already includes shipping, the net margin is understated.
- The data has no explicit Order Status or Payment Status field.
- The dataset covers 2011–2014 only, so the findings describe historical performance.
- Outliers are retained, since they are mostly genuine large orders.

12. Figure Index

All 15 charts are saved in the visualizations folder of the project repository.

No.	File	What it shows
1	01_monthly_sales_trend	Monthly sales with 3-month average
2	02_monthly_net_profit	Monthly net profit
3	03_yearly_sales_profit	Yearly sales, gross and net profit
4	04_seasonality	Average sales by calendar month
5	05_sales_by_category	Sales by category
6	06_category_profit_margin	Net profit and margin by category
7	07_subcategory_net_profit	Net profit by sub-category
8	08_region_sales_vs_profit	Sales and net profit by region
9	09_region_net_margin	Net margin by region
10	10_segment_performance	Customer segment performance
11	11_discount_vs_profit	Discount level vs average net profit
12	12_shipping_cost_by_mode	Shipping cost by ship mode
13	13_sales_distribution	Distribution of sales
14	14_net_profit_distribution	Distribution of net profit
15	15_correlation_heatmap	Correlation heatmap

13. Conclusion

The exploration shows a growing business whose profit is limited by discounting, shipping costs and a few loss-making products and regions. These findings give clear targets for Week 3, where KPIs such as profit margin, cost-to-revenue ratio and product and regional contribution will be calculated formally.